

Ahmed H. Mahmoud

Curriculum Vitae

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RESEARCH EXPERIENCE

Massachusetts Institute of Technology, CSAIL

September 2024 – present

Postdoctoral Associate

- Advisor: Professor Justin Solomon & Professor Jonathan Ragan-Kelley

Autodesk Research, Toronto, Canada

November 2020 – May 2024

Senior Research Scientist

University of California, Davis

March 2016 – October 2020

Graduate Student Researcher

Autodesk Research, Toronto, Canada

June–December 2019, July – November 2020

Intern, Numerical Analysis Research

Shenzhen University, China

June – September 2018

Research intern at the Visual Computing Research Center

EDUCATION

University of California, Davis

June 2024

Ph.D. in Electrical and Computer Engineering

- Advisor: Professor John D. Owens
- Dissertation: Unstructured Geometric Data Processing on the GPU: Data Structures & Programming Models

University of California, Davis

September 2020

M.Sc. in Electrical and Computer Engineering

Alexandria University, Egypt

May 2013

B.S. in Marine Engineering and Naval Architecture

- Very good with honors—ranked first

FUNDING

[1] *OAC Core: OAC Core Projects: GPU Geometric Data Processing*

National Science Foundation (Award # OAC-2403239)

PI. Jonathan Ragan-Kelley, co-PI. Justin Solomon

Amount: **\$600,000**

July 1, 2024–June 30, 2027

Role: The primary author of the technical description

[2] *Efficient GPU Sparse Automatic Differentiation for Scientific Computing*

MIT Generative AI Impact Consortium (MGAIC)

PI. Justin Solomon, co-PI. Jonathan Ragan-Kelley

Amount: **\$150,000**

June 1, 2025–May 31, 2026

Role: The primary author of the technical description

PUBLICATIONS

- [1] *Locality-Aware Automatic Differentiation on the GPU for Mesh-Based Computations*
Ahmed H. Mahmoud, Rahul Goel, Jonathan Ragan-Kelley, and Justin Solomon.
To appear in SIGGRAPH 2026 (arXiv: 2509.00406).
- [2] *Fast Sparse Matrix Permutation for Mesh-Based Direct Solvers*
Behrooz Zarebavani[†], **Ahmed H. Mahmoud**[†], Ana Dodik, Changcheng Yuan, Serban D. Porumbescu, John D. Owens, Maryam Mehri Dehnavi, and Justin Solomon.
To appear in SIGGRAPH 2026 (arXiv: 2602.00898).
[†] joint first author
- [3] *iskra: A System for Inverse Geometry Processing*
Ana Dodik, **Ahmed H. Mahmoud**, and Justin Solomon.
To appear in SIGGRAPH 2026 (arXiv: 2602.12105).
- [4] *Low-Rank Adaptation of Neural Fields*
Anh Truong, **Ahmed H. Mahmoud**, Mina Konaković Luković, and Justin Solomon.
SIGGRAPH Asia 2025.
- [5] *Dynamic Mesh Processing on the GPU*
Ahmed H. Mahmoud, Serban D. Porumbescu, and John D. Owens.
ACM Transactions on Graphics (SIGGRAPH 2025).
- [6] *Disaggregated Design for GPU-Based Volumetric Data Structures*
Massimiliano Meneghin and **Ahmed H. Mahmoud**.
European Conference on Parallel and Distributed Computing (EuroPar 2025)
- [7] *Optimized GPU implementation of grid refinement in lattice Boltzmann method*
Ahmed H. Mahmoud, Hesam Salehipour, and Massimiliano Meneghin
International Parallel and Distributed Processing Symposium (IPDPS 2024)
Open Source Contribution Award
- [8] *Neon: A Multi-GPU Programming Model for Grid-based Computations*
Massimiliano Meneghin[†], **Ahmed H. Mahmoud**[†], Pradeep Kumar Jayaraman, and Nigel J. W. Morris.
International Parallel and Distributed Processing Symposium (IPDPS 2022)
[†] joint first author
- [9] *RXMesh: A GPU Mesh Data Structure*
Ahmed H. Mahmoud, Serban D. Porumbescu, and John D. Owens
ACM Transactions on Graphics (SIGGRAPH 2021)
- [10] *VoroCrust: Voronoi Meshing Without Clipping*
Ahmed Abdelkader, Chandrajit L. Bajaj, Mohamed S. Ebeida, **Ahmed H. Mahmoud**, Scott A. Mitchell, John D. Owens and Ahmad A. Rushdi
ACM Transactions on Graphics (SIGGRAPH 2020)
- [11] *Sampling Conditions for Conforming Voronoi Meshing by the VoroCrust Algorithm*
Ahmed Abdelkader, Chandrajit L. Bajaj, Mohamed S. Ebeida, **Ahmed H. Mahmoud**, Scott A. Mitchell, John D. Owens and Ahmad A. Rushdi
International Symposium on Computational Geometry (SoCG 2018)
- [12] *A Constrained Resampling Strategy for Mesh Improvement*
Ahmed Abdelkader[†], **Ahmed H. Mahmoud**[†], Ahmad A. Rushdi, Scott A. Mitchell, John D. Owens, and Mohamed S. Ebeida
Computer Graphics Forum (presented at ACM/Eurographics Symposium on Geometry Processing SGP 2017)
[†] joint first author
- [13] *All-Quad Meshing without Cleanup*
Ahmad A. Rushdi, Scott A. Mitchell, **Ahmed H. Mahmoud**, Chandrajit L. Bajaj, and Mohamed S. Ebeida
Computer-Aided Design (CAD 2017)

- [14] *Disk Density Tuning of a Maximal Random Packing*
Mohamed S. Ebeida, Ahmad A. Rushdi, Muhammad A. Awad, **Ahmed H. Mahmoud**, Dongming Yan, Shawn English, John D. Owens, Chandrajit L. Bajaj, and Scott A. Mitchell
Computer Graphics Forum (presented at ACM/Eurographics Symposium on Geometry Processing SGP 2016)
- [15] *Exercises in High-Dimensional Sampling: Maximal Poisson-disk Sampling and k-d Darts*
Mohamed S. Ebeida, Scott A. Mitchell, Anjul Patney, Andrew A. Davidson, Stanley Tzeng, Muhammad A. Awad, **Ahmed H. Mahmoud**, and John D. Owens
Book chapter in Topological and Statistical Methods for Complex Data: Tackling Large-Scale, High-Dimensional, and Multi-variate Data Spaces (2014)
- [16] *Delaunay Quadrangulation by Two-coloring Vertices*
Scott A. Mitchell, Mohammed A. Mohammed, **Ahmed H. Mahmoud** and Mohamed S. Ebeida
International Meshing Roundtable (IMR 2014)
- [17] *Improving Spatial Coverage while Preserving the Blue Noise of Point Sets*
Mohamed S. Ebeida, Muhammad A. Awad, Xiaoyin Ge, **Ahmed H. Mahmoud**, Scott A. Mitchell, Patrick M. Knupp, and Li-Yi Wei
SIAM Conference on Geometric and Physical Modeling (SIAM GD/SPM13)
- [18] *Sifted Disks*
Mohamed S. Ebeida, **Ahmed H. Mahmoud**, Muhammad A. Awad, Mohammed A. Mohammed, Scott A. Mitchell, Alex Rand, and John D. Owens
Eurographics 2013.

PATENTS

- [1] *Optimized GPU Implementation of Grid Refinement in the Lattice Boltzmann Method*
Ahmed H. Mahmoud, Hesam Salehipour, and Massimiliano Meneghin
Filed on January 29, 2025 by Autodesk, Inc. [US20250307504A1](#)

SELECTED TALKS

Accelerating Irregular Computation

Queen's University (May 2026 - Kingston, Canada)
University of Toronto (April 2026 - Toronto, Canada)
University of Waterloo (March 2026 - Waterloo, Canada)
University of California, Berkeley (February 2026 - CA, USA)
University of Victoria (November 2025 - Victoria, Canada)

Dynamic Mesh Processing on the GPU

SIGGRAPH (August 2025 - Vancouver, Canada)
Highlights of Parallel Computing (July 2025 - Portland, Oregon)
Brown Visual Computing Seminar (October 2024 - Brown University)
Adobe (November 2023 - Virtual)

RXMesh: A High-performance Mesh Data Structure and Programming Model on the GPU

NVIDIA GTC (March 2022 - Virtual)

Neon: A Multi-GPU Programming Model for Grid-based Computations

NVIDIA GTC (March 2022 - Virtual)

RXMesh: A GPU Mesh Data Structure

SIGGRAPH (August 2021 - Virtual)

A Constrained Resampling Strategy for Mesh Improvement

ACM/Eurographics Symposium on Geometry Processing (July 2017 - London, UK)

MENTEES

Behrooz Zarebavani (Ph.D. Student, University of Toronto) → SIGGRAPH paper

Project: Fast Sparse Matrix Permutation for Mesh-Based Direct Solvers

Ana Dodik (Ph.D. Student, MIT) → SIGGRAPH paper

Project: System for Inverse Geometry Processing

Anh Truong (Ph.D. Student, MIT) → First SIGGRAPH Asia paper

Project: Parameter-efficient Updates of Neural Fields using LoRA

Sachin Kishan (SGI Fellow) → Ph.D. Student at New York University

Project: GPU Geometric Multigrid on Triangle Mesh

Changcheng (Eric) Yuan (M.Sc. Student, UC Davis) → Ph.D. Student, Texas A&M University

Project: GPU Sparse Matrix Reordering

Brooke Dolny (Autodesk Research intern) → M.Sc. Student, University of Waterloo

Project: GPU-accelerated Lattice-Boltzmann Fluid Simulation

TEACHING

Accelerated Computing (6.S894)

Fall 2024, Fall 2025

MIT

Guest Lecturer

Control Systems I (EEC 157A)

Fall 2017

University of California, Davis

Teaching Assistant

Computer Programming (CS224)

Fall 2015

Alexandria University, Egypt

Assistant Lecturer

Ships and Machines Drawing (MR111)

Fall 2015

Alexandria University, Egypt

Assistant Lecturer

Fluid Mechanics (MR231)

Fall 2015

Alexandria University, Egypt

Assistant Lecturer

Fluid Mechanics and Hydraulic Machines (MR232)

Spring 2014

Alexandria University, Egypt

Assistant Lecturer

Marine Hydro-dynamics (OCE323)

Spring 2014

Alexandria University, Egypt

Assistant Lecturer

Theory of Machines (ME145)

Spring 2014

Alexandria University, Egypt

Assistant Lecturer

Material Technology (MR242)

Fall 2013

Alexandria University, Egypt

Assistant Lecturer

Marine Power Plants (MR352)

Fall 2013

Alexandria University, Egypt

Assistant Lecturer

ACADEMIC SERVICE

SIGGRAPH Technical Papers Committee Member	2026
SIGGRAPH Posters Juror	2025, 2026
MIT Summer Geometry Initiative Admission Committee and mentor	2025, 2026
New England Symposium on Graphics Organizing Committee	2025, 2026
ACM/Eurographics Symposium on Geometry Processing Technical Papers Committee	2024, 2025
MIT Summer Geometry Initiative Mentor	2024
High Performance Graphics International Paper Committee	2024
International Conference on Geometric Modeling and Processing Technical Program Committee	2023, 2024
ECE Peer Mentoring Program at UC Davis Mentor	2021, 2023
UC Davis SACNAS's Mentor Match Program Mentor	2023

REFeree SERVICE

ACM Transactions on Architecture and Code Optimization	2026
ACM Transactions on Parallel Computing	2026
SIGGRAPH	2024, 2025
SIGGRAPH Asia	2024
Computers & Graphics	2024
Transactions on Visualization and Computer Graphics	2023
Eurographics	2023
Computer Aided Geometric Design	2022
The SIAM International Meshing Roundtable Workshop	2022, 2023, 2024
International Meshing Roundtable	2019, 2021
Computer-Aided Design	2019

MEDIA COVERAGE

Sandia LabNews Automating complex 3D modeling (webpage , pdf)	April 2020
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REFERENCES

John D. Owens
Child Family Professor of Engineering and Entrepreneurship – University of California, Davis
jowens@ece.ucdavis.edu

Justin Solomon

Associate Professor – Massachusetts Institute of Technology

jsolomon@mit.edu

Jonathan Ragan-Kelley

Associate Professor – Massachusetts Institute of Technology

jrk@mit.edu

Mohamed S. Ebeida

Founder and CEO – Discreetize. Previously, a Principal Member Of Technical Staff – Sandia National Laboratories

msebeida@gmail.com